

SAN LUIS VALLEY, CO, USA

WMO: 724620

Lat: 37.439N

Lon: 105.861W

Elev: 7533

StdP: 11.11

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 23061

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.6	-7.8	-18.7	2.6	-11.6	-13.1	3.6	-3.1	27.6	37.1	24.5	35.7	2.8	260	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.6	85.9	55.0	83.7	54.6	81.4	54.4	59.4	76.0	58.5	74.6	57.5	73.1	10.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.1	85.7	61.1	53.6	81.2	61.0	52.2	77.0	61.2	29.4	76.0	28.6	74.6	27.9	72.8	65.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.1	24.0	20.2	DB	-20.5	90.2	8.2	1.9	-26.4	91.5	-31.2	92.6	-35.8	93.7	-41.8	95.0	(4)
(5)			WB	-20.8	61.8	8.0	1.4	-26.6	62.8	-31.3	63.6	-35.8	64.3	-41.6	65.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.7	17.9	24.7	35.3	42.5	51.6	60.7	65.3	63.0	55.9	43.9	30.9	19.3		
	DBStd	17.78	9.87	9.44	7.52	6.55	6.33	4.49	3.19	3.09	5.26	6.88	8.48	9.57		
	HDD50	4197	994	707	455	233	59	1	0	0	14	209	572	953		
	HDD65	8215	1459	1127	920	675	416	139	34	76	274	655	1022	1418		
	CDD50	1527	0	0	0	8	108	322	476	403	191	19	0	0		
	CDD65	70	0	0	0	0	0	10	45	14	1	0	0	0		
	CDH74	3309	0	0	0	1	139	956	1227	740	242	4	0	0		
	CDH80	768	0	0	0	0	16	243	351	136	22	0	0	0		
	WSAvg	7.2	5.2	6.8	8.4	10.2	9.6	8.7	6.6	6.1	6.5	6.7	5.9	5.2		
	PrecAvg	7.5	0.2	0.3	0.5	0.5	0.6	0.6	1.1	1.2	0.9	0.7	0.4	0.4		
	PrecMax	11.5	0.7	1.4	1.6	1.7	1.9	2.6	3.5	3.3	1.9	2.4	1.2	1.5		
(15) Precipitation	PrecMin	4.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0		
	PrecStd	1.8	0.2	0.3	0.4	0.4	0.5	0.5	0.7	0.7	0.5	0.6	0.4	0.4		
	DB	51.3	56.8	67.6	72.4	82.3	89.1	89.3	86.1	82.8	74.2	63.2	52.5			
	MCWB	36.2	38.6	42.9	45.3	51.5	54.3	56.8	55.2	54.2	48.5	41.9	37.1			
	2% DB	46.0	52.0	62.8	68.4	77.8	85.4	86.5	83.6	79.5	70.7	59.1	47.2			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	33.4	36.4	40.8	43.5	48.9	53.2	56.1	55.4	52.7	47.0	40.6	34.5			
	5% DB	41.5	47.6	58.8	65.4	74.4	82.9	84.1	81.3	76.8	67.1	54.7	42.6			
	MCWB	31.1	34.4	39.1	42.3	47.6	52.2	56.0	55.5	51.7	45.7	38.6	32.1			
	10% DB	36.8	42.7	54.3	62.2	71.3	80.1	81.4	78.7	73.9	63.2	50.0	37.7			
	MCWB	29.0	32.0	37.3	41.2	46.4	51.4	56.2	55.4	51.2	44.1	36.2	29.3			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	37.2	39.9	43.6	47.0	53.2	58.3	61.2	60.6	57.7	52.5	43.9	38.6			
	MCDB	48.8	53.6	64.3	66.5	75.4	77.2	76.7	76.9	73.6	66.6	58.6	49.3			
	2% WB	34.2	37.2	41.5	45.0	50.9	56.5	59.8	59.3	56.0	49.7	41.5	35.4			
	MCDB	43.6	49.9	60.2	64.6	71.8	75.7	76.7	74.7	71.4	64.3	56.8	45.6			
	5% WB	32.0	34.9	39.8	43.4	49.3	55.0	59.0	58.2	54.6	47.4	39.2	32.5			
(31) Mean Daily Temperature Range	MCDB	39.9	46.2	56.8	62.3	68.8	74.8	75.9	73.2	69.9	63.0	52.8	41.4			
	10% WB	29.6	32.6	37.8	41.8	47.8	53.5	58.1	57.2	53.2	45.3	36.9	29.8			
	MCDB	36.9	41.6	52.8	60.2	67.4	73.4	74.4	71.8	68.2	60.6	49.1	37.9			
	5% DB	33.7	31.8	33.2	32.8	34.2	37.4	33.6	32.7	34.9	35.5	34.7	32.7			
	MCDBR	38.2	36.8	40.5	39.9	40.0	42.2	38.1	37.2	39.8	41.2	41.2	37.2			
(36) Clear-Sky Solar Irradiance	MCWBR	26.7	22.9	21.4	19.3	16.5	15.9	13.1	13.7	16.6	20.9	24.5	25.3			
	5% WB	35.1	34.2	37.6	37.0	35.6	36.2	31.2	30.0	31.6	36.4	38.5	34.8			
	MCWBR	25.0	21.5	20.2	18.3	14.7	13.8	11.1	11.3	13.6	19.4	23.2	24.0			
	taub	0.221	0.229	0.260	0.278	0.297	0.302	0.348	0.325	0.286	0.256	0.235	0.226			
	taud	2.615	2.607	2.531	2.508	2.507	2.491	2.427	2.464	2.593	2.651	2.627	2.591			
(42) All-Sky Solar Radiation	Ebn at Noon	319	327	322	318	312	308	293	299	309	312	311	310			
	Edh at Noon	22	25	30	33	34	34	36	34	28	24	22	21			
	RadAvg	980	1312	1674	2003	2244	2422	2120	1948	1756	1389	1043	846			
(45)	RadStd	53	85	115	113	161	126	107	84	93	83	60	46			

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+95	+18	(46)
(47)	Regional (21 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+146	+70	(47)

Nomenclature: See separate page